UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: ARTIFICIAL INTELLEGENCE

SUBMITTED BY & TO

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REG NO : 23MDBCS 435

SECTION : A

SEMESTER : 4th

SUBMITTED T0 : MAM FAIZA

LAB TASK # 1

TASK 1

Temperatue converter: write a program that converts a temperatue from fahrenheit to celcius.

CODE

task 1.1.py - E:/University/SEMESTER 4th/programs/task 1.1.py (3.13.1)

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```
fahrenheit = float(input("Enter temperature in Fahrenheit: "))
celsius = (fahrenheit - 32) * 5 / 9
print(f"Temperature in Celsius: {celsius:.2f}")
```

OUTPUT

IDLE Shell 3.13.1

File Edit Shell Debug Options Window Help

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=========== RESTART: E:/University/SEMESTER 4th/programs/task 1.1.py ===========
Enter temperature in Fahrenheit: 45
Temperature in Celsius: 7.22
>>>
```

TASK 2

Grade calculator: write a program that calculates a student's grade base on their score.

untitled

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```python
score = int(input("Enter your score (0-100): "))

if 90 <= score <= 100:
    grade = 'A'
elif 80 <= score < 90:
    grade = 'B'
elif 70 <= score < 80:
    grade = 'C'
elif 60 <= score < 70:
    grade = 'D'
elif 0 <= score < 60:
    grade = 'F'
else:
    print("Invalid score.")
    exit()

print(f"Your grade is: {grade}")
```

OUTPUT

IDLE Shell 3.13.1

File Edit Shell Debug Options Window Help

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=========== RESTART: E:/University/SEMESTER 4th/programs/task 1.2.py ===========
Enter your score (0-100): 89
Your grade is: B
>>>
```

TASK 3

FizzBuzz: Write a program that prints the numbers from 1 to 100, but for multiples of 3 print

"Fizz" instead of the number, and for multiples of 5 print "Buzz". For numbers which are

multiples of both 3 and 5, print "FizzBuzz".

CODE

untitled

File Edit Format Run Options Window Help

```
for i in range(1, 101):
    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")
    elif i % 3 == 0:
        print("Fizz")
    elif i % 5 == 0:
        print("Buzz")
    else:
        print(i)
```

OUTPUT

IDLE Shell 3.13.1

File Edit Shell Debug Options Window Help

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28)
Type "help", "copyright", "credits" or "license()" for more
>>>
=========== RESTART: E:/University/SEMESTER 4th/programs/tas
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
```

TASK 4

Sum of Numbers: Write a program that calculates the sum of the numbers from 1 to 100.

untitled

File Edit Format Run Options Window Help

```python
total_sum = 0
for i in range(1, 101):
    total_sum += i
print("The sum is:", total_sum)
```

OUTPUT

IDLE Shell 3.13.1

File Edit Shell Debug Options Window Help

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:/University/SEMESTER 4th/programs/task 1.4.py ==========
The sum is: 5050
>>>
```